Official College Board Digital SAT Question Bank — Released Questions

Predicted for the August SAT — CLC Educational Institute

Algebra

Math | 23 questions with full Rule / Tip / Trap / POE / Desmos explanations

The CLC Method

Every question in this file is broken down the same way, so you build one consistent habit instead of guessing question to question:

The Rule — the underlying math principle or formula being tested. Learn the rule, not just this one answer, and you'll be able to answer every future question of this type.

The Tip — a concrete move you can make before diving into algebra, so you solve efficiently instead of the long way.

Step-by-Step Solution — every single algebra step written out in full, with nothing skipped — so you can follow exactly how the answer was reached, not just see the final number.

The Trap — the single wrong answer most students actually pick, and the specific arithmetic or logical slip that produces it.

POE (multiple-choice questions only) — a full walkthrough of the other wrong choices, showing exactly what error produces each one. Some Math questions are Student-Produced Response ('grid-in') questions with no answer choices at all — for those, there's no POE section, just the full solution.

Solving with Desmos — exact, specific instructions for using the built-in Desmos graphing calculator (available on every Digital SAT Math question) to solve or verify the question — including the exact expressions to type in, not just a general suggestion to 'graph it.'

Linear equations in one variable

5 questions · Easy 2 / Medium 2 / Hard 1

Linear equations in one variable — Q1 [Easy] (ID: 5eb4918d)

Question ID: 5eb4918d

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear equations in one variable	Easy

Question

If $9x + 4 = 67$, what is the value of $90x + 40$?

Answer

A. 7

B. 70

C. 130

D. 670

Correct Answer: D

The Rule: This is a substitution shortcut problem. When a question gives you one equation and asks for the value of a MULTIPLE of that same expression (here, 90x+40 is exactly 10 times 9x+4), you almost never need to solve for x itself — you can scale the whole original equation up or down instead.

The Tip: Before solving for x the long way, look closely at the expression you're asked to find (90x+40) and compare it to the expression you're given (9x+4=67). Notice that 90x is exactly 10 times 9x, and 40 is exactly 10 times 4. That means you can just multiply the entire original equation by 10, instead of solving for x and then plugging it back in — this saves an entire step and avoids fraction errors.

Why D is right: Multiply both sides of $9x + 4 = 67$ by 10: this gives $90x + 40 = 670$. So without ever finding x by itself, we directly get the value of $90x + 40$, which is 670.

Step-by-Step Solution:

Step 1: Write down the given equation: $9x + 4 = 67$.

Step 2: Notice $90x + 40 = 10 \times (9x + 4)$.

Step 3: Multiply both sides of the original equation by 10: $10(9x + 4) = 10(67)$, which gives $90x + 40 = 670$.

Step 4: The answer is 670, matching choice D.

(For reference, the long way also works: $9x + 4 = 67 \rightarrow 9x = 63 \rightarrow x = 7$. Then $90(7) + 40 = 630 + 40 = 670$ — same answer, just slower.)

The Trap: Choice A (7) is the most commonly picked wrong answer — students solve correctly for $x = 7$, then stop there and select 7 as their final answer, forgetting that the question never asked for x — it asked for the value of $90x + 40$, a completely different quantity.

POE — Eliminate the other three:

✗ **A —** 7 is the value of x itself, not the value of $90x + 40$ that the question actually asks for — a classic 'answered the wrong question' trap.

✗ **B —** 70 comes from a common arithmetic slip — for example, mistakenly computing 10x instead of $90x + 40$, or dividing instead of multiplying somewhere in the process.

✗ **C —** 130 doesn't correspond to any correct or nearly-correct path through this problem — it's a distractor value with no clear derivation from the given equation.

Solving with Desmos:

Step 1: Open the Desmos graphing calculator (or the built-in graphing calculator in Bluebook during the actual exam).

Step 2: Click on the first empty input row and type the original equation exactly as written: 9x+4=67

Step 3: Press Enter. Because this is a one-variable equation, Desmos will automatically plot the solution as a labeled point on the x-axis — it will show the point (7, 0), telling you $x = 7$.

Step 4: Click on the next empty row and type the target expression using that x-value directly: 90*7+40

Step 5: Press Enter — Desmos will instantly evaluate this and display 670 as the result, which is your final answer, matching choice D.

Step 6 (shortcut for future problems like this): You can skip Steps 3–4 entirely by just typing 10(9x+4) in one row and 10*67 in another to confirm they're equal, or by typing 90x+40 and 670 as two separate horizontal lines on the same graph — if they're the same value, the lines will coincide perfectly at $x = 7$.

Linear equations in one variable — Q2 [Easy] (ID: 935f3403)

Question ID: 935f3403

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear equations in one variable	Easy

Question

For a restaurant owner to purchase a pizza oven at a total price of $13,000$ dollars, a onetime down payment is required, and then fixed monthly payments are made for the remaining amount owed for the pizza oven. The equation $13,000 = 2,800 + 200t$ represents this situation, where t is the number of fixed monthly payments that are made. Which of the following is the best interpretation of 200 in this context?

Answer

A. The amount, in dollars, of the down payment

B. The amount, in dollars, of each fixed monthly payment

C. The total amount, in dollars, paid for the pizza oven after t fixed monthly payments

D. The total number of fixed monthly payments

Correct Answer: B

The Rule: In a real-world linear equation of the form (total) = (fixed amount) + (rate)(number of units), the RATE — the number multiplied by the variable — always represents 'how much per one unit,' while the constant by itself represents the one-time fixed amount.

The Tip: Whenever a word problem gives you an equation like 13,000 = 2,800 + 200t and asks what a specific number in it 'means,' locate that number in the equation first, then ask: 'Is this number sitting alone (a fixed one-time amount), or is it being multiplied by the variable (a rate per unit)?' The answer choices will usually offer one of each type as a trap.

Why B is right: In the equation 13,000 = 2,800 + 200t, the number 200 is multiplied by t (the number of monthly payments) — so 200 represents the dollar amount added for EACH additional monthly payment, i.e., the amount of each fixed monthly payment.

Step-by-Step Solution:

Step 1: Identify what t represents: the number of monthly payments made (given directly in the problem).

Step 2: Look at the structure of the equation: 13,000 = 2,800 + 200t. This matches the general pattern (total price) = (down payment) + (payment amount) × (number of payments).

Step 3: Since 200 is being multiplied by t, and t counts monthly payments, 200 must be the dollar amount of each individual monthly payment.

Step 4: Match this interpretation to the choices: 'the amount, in dollars, of each fixed monthly payment' is choice B.

The Trap: Choice A is the most commonly picked wrong answer — students sometimes attach '200' to the wrong part of the story, confusing it with the down payment, when the down payment is actually the standalone constant 2,800, not the number being multiplied by t.

POE — Eliminate the other three:

✗ A — The down payment is the STANDALONE number in the equation (2,800), not the number multiplied by t — 200 is not the down payment.

✗ C — The total amount paid after t payments would be the ENTIRE right-hand side of the equation (2,800 + 200t), not just the coefficient 200 by itself.

✗ D — The total number of fixed monthly payments is what t itself represents (the variable), not the number 200 that multiplies it.

Solving with Desmos:

This particular question is about interpreting what a number MEANS in a real-world context, not about calculating an unknown value — so Desmos isn't needed to find the answer directly. However, you can use Desmos to double check your understanding of the relationship: Step 1: Type y=2800+200x into Desmos. Step 2: Use the table feature (click the '+' button, then choose the table icon, or just add a table below your equation) and enter x = 0, 1, 2, 3 in the x-column. Step 3: Look at how y changes each time x increases by 1 — you'll see y jumps by exactly 200 each time (2800, 3000, 3200, 3400...). This confirms 200 is the amount added per additional payment (per unit of t), verifying that 200 represents each monthly payment's dollar amount, matching choice B.

Linear equations in one variable — Q3 [Medium] (ID: 088bc84e)

Question ID: 088bc84e

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear equations in one variable	Medium

Question

If $2(5x) = 6$, what is the value of $5x$?

Answer

A. $\frac{4}{5}$

B. $\frac{5}{3}$

C. 3

D. 4

Correct Answer: C

The Rule: When a number multiplies a full parenthetical expression, like 2(5x) = 6, you can treat the parenthetical expression (5x) as a single unit and divide both sides by the OUTSIDE number to isolate it directly — without ever solving for x by itself first.

The Tip: Whenever the question asks for the value of an expression that's already sitting inside parentheses in the given equation (here, 5x is already grouped together), don't break it apart to solve for the single variable first — divide to isolate that exact grouped expression instead. This is faster and avoids unnecessary extra steps.

Why C is right: Starting from 2(5x) = 6, divide both sides by 2 to isolate the grouped expression (5x) directly: 5x = 6 ÷ 2 = 3. The question asks for the value of 5x, which is 3.

Step-by-Step Solution:

Step 1: Start with the given equation: 2(5x) = 6.

Step 2: Divide both sides by 2 to isolate the grouped expression (5x): 5x = 6 ÷ 2 = 3.

Step 3: The value of 5x is 3, matching choice C.

The Trap: Choice B (5/3) is the most commonly picked wrong answer — it comes from solving all the way down to x by itself (5x = 3, so x = 3/5) and then accidentally flipping that fraction upside down, or from confusing 'the value of 5x' with 'the value of x' partway through.

POE — Eliminate the other three:

✗ A — 4/5 doesn't correspond to a natural error path from this equation — it's a distractor value close in form to the other fraction choice.

✗ B — 5/3 typically comes from solving for x alone (x = 3/5) and then flipping the fraction by mistake, or from dividing 5 by 3 instead of correctly isolating 5x as a single unit.

✗ D — 4 doesn't correspond to a natural error path from this equation either — likely a distractor value with no clear derivation.

Solving with Desmos:

Step 1: Open Desmos and type the equation exactly as given: 2(5x)=6

Step 2: Press Enter — Desmos will solve this one-variable equation and plot the solution as a labeled point on the x-axis, showing the value of x directly (x = 0.6, or 3/5).

Step 3: Since the question asks for 5x specifically (not just x), type a second line: 5*0.6 — Desmos will instantly evaluate this and display 3, confirming the value of 5x is 3, matching choice C.

Step 4: Alternatively, skip solving for x entirely — just type 6/2 directly into Desmos, since dividing both sides of 2(5x)=6 by 2 gives 5x = 6/2 = 3 immediately.

Linear equations in one variable — Q4 [Medium] [Grid-In] (ID: f8fa0cf0)

Question ID: f8fa0cf0

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear equations in one variable	Medium

Question

$$9r = 7(r + 7)$$

What value of r is the solution to the given equation?

Correct Answer: 49/2 or 24.5 (Student-Produced Response — no answer choices given)

The Rule: This is a straightforward one-variable linear equation with a distributed term. The rule is: always distribute (multiply through the parentheses) FIRST, then collect all the variable terms on one side and all the constant terms on the other, before dividing to isolate the variable.

The Tip: When you see a variable on both sides of the equation after distributing (like 9r on the left and 7r on the right here), move all r-terms to one side by subtracting the smaller coefficient's term from both sides — this keeps your remaining coefficient positive and avoids sign errors.

Why this is the answer: Starting from 9r = 7(r + 7), distribute the 7 on the right side: 9r = 7r + 49. Subtract 7r from both sides: 2r = 49. Divide both sides by 2: r = 49/2, or equivalently 24.5.

Step-by-Step Solution:

Step 1: Write the given equation: 9r = 7(r + 7).

Step 2: Distribute the 7 across the parentheses on the right side: 9r = 7r + 49.

Step 3: Subtract 7r from both sides to collect the r-terms on the left: 9r − 7r = 49, which simplifies to 2r = 49.

Step 4: Divide both sides by 2 to isolate r: r = 49/2 = 24.5.

Step 5: This is a grid-in (student-produced response) question, so you would enter either 49/2 or 24.5 into the answer grid — both are accepted as correct.

The Trap: A common error on this type of problem is distributing incorrectly — for example, writing 7(r+7) as 7r + 7 (forgetting to multiply the 7 by BOTH terms inside the parentheses), which would lead to the wrong equation 9r = 7r + 7 and an incorrect final answer of r = 3.5.

Solving with Desmos:

Step 1: Open Desmos and type the equation exactly as given, using r as the variable: 9r=7(r+7)

Step 2: Press Enter. Since this is a one-variable equation, Desmos will solve it automatically and plot the solution as a labeled point on the r-axis (which will display as the x-axis in Desmos, since Desmos treats whatever letter you use as its horizontal variable).

Step 3: Click directly on the labeled point to see its exact coordinates — it will show r = 24.5 (which Desmos may also display as the fraction 49/2 if you click to toggle the coordinate display format).

Step 4: Enter 24.5 or 49/2 into the grid-in answer box on the actual test — both forms are accepted.

Linear equations in one variable — Q5 **[Hard]** (ID: 55ea0659)

Question ID: 55ea0659

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear equations in one variable	Hard

Question

$$x(r-9)+4=19x+27$$

In the given equation, r is a positive integer. If the given equation has exactly one solution, what CANNOT be the value of r?

Answer

A. 4

B. 9

C. 23

D. 28

Correct Answer: D

The Rule: For an equation of the form x(r − a) = bx + c to have exactly ONE solution, the coefficient of x after simplifying must NOT be zero. If the coefficient becomes zero, the equation either has NO solution (if the constants don't match) or INFINITELY MANY solutions (if they do) — never exactly one. So the question is really asking: for what value of r does the x-coefficient become zero?

The Tip: Whenever a question asks what value 'CANNOT' make an equation have exactly one solution, your first move should be to simplify the equation into the form (coefficient)x = (constant), then figure out what makes that coefficient equal to zero — that specific value is almost always the 'cannot be' answer.

Why D is right: Starting from x(r − 9) + 4 = 19x + 27, distribute and rearrange: xr − 9x + 4 = 19x + 27 → xr − 9x − 19x = 27 − 4 → xr − 28x = 23 → x(r − 28) = 23. This equation has exactly one solution UNLESS the coefficient (r − 28) equals zero — which happens when r = 28. If r = 28, the equation becomes 0 = 23, which is never true (no solution, not one solution). So r cannot equal 28.

Step-by-Step Solution:

Step 1: Start with x(r − 9) + 4 = 19x + 27.

Step 2: Distribute x across (r − 9): xr − 9x + 4 = 19x + 27.

Step 3: Move all x-terms to the left side and constants to the right: xr − 9x − 19x = 27 − 4.

Step 4: Combine like terms: xr − 28x = 23, which factors to x(r − 28) = 23.

Step 5: For this equation to have exactly one solution, the coefficient (r − 28) must NOT be zero — because if it were zero, the left side would be 0, and 0 = 23 is impossible (no solution at all, not one).

Step 6: Set r − 28 = 0 to find the forbidden value: r = 28. This is the value r cannot be, matching choice D.

The Trap: Choice A (4) is the most commonly picked wrong answer for students who correctly simplify to x(r − 28) = 23 but then make an arithmetic slip while solving for the forbidden r-value, perhaps confusing which number to set the coefficient equal to zero against.

POE — Eliminate the other three:

✗ A — 4 does not make the coefficient (r − 28) equal zero — plugging r = 4 gives x(4 − 28) = 23, or −24x = 23, which has exactly one clean solution, so r = 4 is perfectly fine.

✗ B — 9 does not make (r − 28) equal zero either — this value doesn't create any problem with the equation having one solution.

✗ C — 23 also does not make (r − 28) equal zero — like the other wrong choices, this value still allows exactly one solution.

Solving with Desmos:

Step 1: Open Desmos. Since this problem involves TWO variables (x and r) and asks about a special condition, direct equation-solving won't show the answer as neatly as it does for simpler problems — but you can still verify your algebra.

Step 2: First, simplify the equation by hand (or using Desmos as a calculator) to the form $x(r - 28) = 23$, as shown in the steps above. Step 3: To CONFIRM $r = 28$ breaks the equation, type this test directly into Desmos as a check: x*(28-28)=23 — Desmos will show this simplifies to $0 = 23$, which Desmos will indicate has no solution (no point will plot), confirming $r = 28$ is impossible. Step 4: To confirm any of the wrong answers (like $r = 4$) DO work, type x*(4-28)=23 into Desmos — it will solve this immediately and plot a valid point, confirming $r = 4$ is a value that r CAN be, ruling it out as the answer.

Linear equations in two variables

2 questions · Easy 0 / Medium 1 / Hard 1

Linear equations in two variables — Q1 [Medium] (ID: 4722abd2)

Question ID: 4722abd2

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear equations in two variables	Medium

Question

Line p is defined by $2y + 8x = 11$. Line r is perpendicular to line p in the *xy*-plane. What is the slope of line r?

Answer

A. -4

B. $-\frac{1}{4}$

C. $\frac{1}{4}$

D. 4

Correct Answer: C

The Rule: For two lines to be perpendicular in the xy-plane, their slopes must be negative reciprocals of each other — meaning if one line has slope m, the perpendicular line has slope $-1/m$. Always rewrite a given equation into slope-intercept form ($y = mx + b$) first to clearly identify its slope before finding a perpendicular slope.

The Tip: When an equation is given in a mixed-up form like $2y + 8x = 11$, always isolate y completely first (getting it into $y = mx + b$ form) before trying to identify the slope — trying to read the slope directly off the unsimplified equation is a common source of sign errors.

Why C is right: Rewrite $2y + 8x = 11$ by isolating y: $2y = -8x + 11$, then divide everything by 2: $y = -4x + 11/2$. This shows line p has a slope of -4. A line perpendicular to p must have a slope equal to the negative reciprocal of -4, which is $1/4$.

Step-by-Step Solution:

Step 1: Start with the given equation for line p: $2y + 8x = 11$.

Step 2: Subtract 8x from both sides: $2y = -8x + 11$.

Step 3: Divide every term by 2 to fully isolate y: $y = -4x + 11/2$.

Step 4: Identify the slope of line p from this slope-intercept form: the slope is -4 (the coefficient of x).

Step 5: To find the slope of a line perpendicular to p, take the negative reciprocal of -4. Flip the fraction ($-4 = -4/1$ becomes $-1/4$) and then flip the sign: the negative reciprocal is $1/4$.

Step 6: The slope of line r is $1/4$, matching choice C.

The Trap: Choice A (-4) is the most commonly picked wrong answer — students correctly find the slope of line p but then forget the question asks for the PERPENDICULAR line's slope, mistakenly selecting line p's own slope instead of its negative reciprocal.

POE — Eliminate the other three:

✗ **A —** −4 is the slope of line p itself, not the slope of the perpendicular line r that the question actually asks for.

✗ **B —** −1/4 is the negative of the reciprocal's sign but keeps the same sign as the original slope — this comes from only flipping the fraction without also flipping the sign, an incomplete negative-reciprocal calculation.

✗ **D —** 4 comes from taking the reciprocal (flipping the fraction) but forgetting to change the sign — since the original slope is negative, the perpendicular slope must be positive, but the magnitude here (4 instead of 1/4) is also flipped incorrectly.

Solving with Desmos:

Step 1: Open Desmos and type the original equation for line p exactly as given: 2y+8x=11

Step 2: Press Enter — Desmos will graph the line automatically, and you can visually confirm it's a steep, downward-sloping line.

Step 3: Click on the line itself, or hover over it, and Desmos may show you its slope directly, or you can click the wrench/settings icon to view its equation in slope-intercept form.

Step 4: To find and verify the perpendicular slope, type a new line using the negative reciprocal you calculated: y=(1/4)x+0 (the y-intercept doesn't matter for checking perpendicularity, just the slope).

Step 5: Visually confirm on the graph that these two lines cross at a perfect right angle (90 degrees) — Desmos graphs are drawn to scale, so a true perpendicular intersection will visibly look like a plus sign (+) where the lines cross, confirming slope 1/4 is correct.

Linear equations in two variables — Q2 [Hard] [Grid-In] (ID: 107d0c62)

Question ID: 107d0c62

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear equations in two variables	Hard

Question

$$0.10x + 0.20y = 0.17\,(x + y)$$

The equation gives a volume x, in gallons, of a 10% solution that could be mixed with a volume y, in gallons, of a 20% solution to produce a 17% solution. According to this equation, what volume, in gallons, of the 20% solution could be mixed with 90.0 gallons of the 10% solution to produce a 17% solution? (Assume that the volume of the mixture is the sum of the volumes of the two solutions before they were mixed.)

Correct Answer: 210 (Student-Produced Response — no answer choices given)

The Rule: Mixture problems give you a general equation relating two unknown quantities (here, x and y). When the problem later tells you a SPECIFIC value for one variable, substitute that specific number in for that variable FIRST, then solve the resulting one-variable equation for the other unknown.

The Tip: Don't try to solve a mixture equation with two unknowns in general form — as soon as the problem gives you a specific number for one of the variables (here, 'mixed with 90.0 gallons of the 10% solution' tells you x = 90), plug that number in immediately to reduce the problem to a simple one-variable equation.

Why this is the answer: Substitute x = 90 into the equation 0.10x + 0.20y = 0.17(x + y): 0.10(90) + 0.20y = 0.17(90 + y). This simplifies to 9 + 0.20y = 15.3 + 0.17y. Subtracting 0.17y from both sides and 9 from both sides gives 0.03y = 6.3, so y = 6.3 / 0.03 = 210.

Step-by-Step Solution:

Step 1: Start with the given equation: 0.10x + 0.20y = 0.17(x + y).

Step 2: Substitute the known value x = 90 into the equation: 0.10(90) + 0.20y = 0.17(90 + y).

Step 3: Simplify the left side: 0.10 × 90 = 9, so the left side becomes 9 + 0.20y.

Step 4: Distribute the 0.17 on the right side: 0.17 × 90 = 15.3, and 0.17 × y = 0.17y, so the right side becomes 15.3 + 0.17y.

Step 5: Now the equation reads: 9 + 0.20y = 15.3 + 0.17y.

Step 6: Subtract 0.17y from both sides: 9 + 0.03y = 15.3.

Step 7: Subtract 9 from both sides: 0.03y = 6.3.

Step 8: Divide both sides by 0.03: y = 6.3 / 0.03 = 210.

Step 9: This is a grid-in question — enter 210 as your final answer, representing 210 gallons of the 20% solution.

The Trap: A common error on this problem is forgetting to distribute the 0.17 across BOTH terms inside the parentheses (90 + y) — students sometimes only multiply 0.17 by y and forget to also multiply it by 90, leading to an incorrect final answer.

Solving with Desmos:

Step 1: Open Desmos. Step 2: Type the original equation, substituting x = 90 directly: 0.10(90)+0.20y=0.17(90+y)

Step 3: Press Enter — since y is now the only variable, Desmos will solve this one-variable equation automatically and plot the solution as a labeled point on the y-axis.

Step 4: Click on the labeled point to see its exact coordinates — it will display y = 210, confirming your answer.

Step 5: Enter 210 into the grid-in answer box on the actual test.

Systems of two linear equations in two variables

7 questions · Easy 1 / Medium 3 / Hard 3

Systems of two linear equations in two variables — Q1 **[Easy]** (ID: 0e926898)

Question ID: 0e926898

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Systems of two linear equations in two variables	Easy

Question

Students and chaperones from a middle school are going on a field trip to a museum. Admission to the museum will cost $\$8.50$ for each student and $\$12$ for each chaperone. It will cost a total of $\$730$ for admission to the museum for all 83 people on the field trip. How many students are going on the field trip?

Answer

A. 31

B. 70

C. 76

D. 86

Correct Answer: C

The Rule: Word problems describing two different groups (students and chaperones) with a combined total count and a combined total cost always translate into a SYSTEM of two equations: one equation for the total COUNT of people, and one equation for the total COST. Solve the system using substitution or elimination.

The Tip: When setting up a system from a word problem, write out both equations in words first before converting to symbols: 'number of students + number of chaperones = total people' and 'cost per student × students + cost per chaperone × chaperones = total cost.' This helps prevent mixing up which number goes where.

Why C is right: Let s = number of students and c = number of chaperones. The two equations are: s + c = 83 (total people) and 8.50s + 12c = 730 (total cost). Solving this system by substitution: from the first equation, c = 83 − s. Substitute into the second equation: 8.50s + 12(83 − s) = 730, which simplifies to 8.50s + 996 − 12s = 730, then −3.5s = −266, so s = 76.

Step-by-Step Solution:

Step 1: Define variables: let s = number of students, c = number of chaperones.

Step 2: Write the total-people equation: $s + c = 83$.

Step 3: Write the total-cost equation: $8.50s + 12c = 730$.

Step 4: Solve the first equation for c: $c = 83 - s$.

Step 5: Substitute this expression for c into the second equation: $8.50s + 12(83 - s) = 730$.

Step 6: Distribute the 12: $8.50s + 996 - 12s = 730$.

Step 7: Combine the s-terms: $8.50s - 12s = -3.5s$, so the equation becomes $-3.5s + 996 = 730$.

Step 8: Subtract 996 from both sides: $-3.5s = -266$.

Step 9: Divide both sides by -3.5: $s = 76$.

Step 10: This matches choice C — 76 students are going on the field trip.

The Trap: Choice D (86) is the most commonly picked wrong answer — it often results from a sign error while distributing the 12 across $(83 - s)$, such as writing $12(83 - s)$ as $996 + 12s$ instead of the correct $996 - 12s$, which flips the direction of the final subtraction.

POE — Eliminate the other three:

✗ A — 31 would be the value of c (the number of chaperones), not s (the number of students) — this is likely selected by students who solve the system correctly but grab the wrong variable's value as their final answer.

✗ B — 70 doesn't correspond to a clean derivation from this system — likely an arithmetic slip somewhere in the distribution or combination step.

✗ D — 86 typically results from a sign error while distributing 12 across $(83 - s)$, flipping a subtraction into an addition partway through the solution.

Solving with Desmos:

Step 1: Open Desmos. Step 2: Type the first equation exactly as derived: s+c=83 — but since Desmos uses x and y by default for graphing two-variable equations, it's easiest to substitute x for s and y for c: x+y=83

Step 3: Press Enter — Desmos will graph this as a line.

Step 4: Click on the next empty row and type the second equation, again using x for students and y for chaperones: 8.50x+12y=730

Step 5: Press Enter — Desmos will graph this second line as well.

Step 6: Look for the point where the two lines cross (intersect) — this is the solution to the system. Click directly on that intersection point, and Desmos will display its exact coordinates.

Step 7: The x-coordinate of that intersection point is the number of students (76), and the y-coordinate is the number of chaperones (7) — confirming s = 76 matches choice C.

Systems of two linear equations in two variables — Q2 [Medium] (ID: ea8bd66f)

Question ID: ea8bd66f

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Systems of two linear equations in two variables	Medium

Question

$$y = 4x$$

$$5x - 3y = -14$$

The solution to the given system of equations is (x, y). What is the value of $x + y$?

Answer

A. 4

B. 6

C. 8

D. 10

Correct Answer: D

The Rule: When one equation in a system is already solved for one variable (like y = 4x), the fastest solving method is SUBSTITUTION — plug that expression directly into the other equation in place of that variable, turning the system into a single one-variable equation.

The Tip: Whenever you see a system where one equation looks like 'y = (something with x)' or 'x = (something with y),' that's your signal to use substitution rather than elimination — it's almost always faster in that specific setup.

Why D is right: Since y = 4x, substitute 4x in place of y in the second equation: 5x − 3(4x) = −14. This simplifies to 5x − 12x = −14, so −7x = −14, giving x = 2. Then y = 4x = 4(2) = 8. Finally, x + y = 2 + 8 = 10.

Step-by-Step Solution:

Step 1: Note that the first equation is already solved for y: y = 4x.

Step 2: Substitute 4x in place of y into the second equation: 5x − 3(4x) = −14.

Step 3: Distribute the −3: 5x − 12x = −14.

Step 4: Combine like terms: −7x = −14.

Step 5: Divide both sides by −7: x = 2.

Step 6: Substitute x = 2 back into y = 4x to find y: y = 4(2) = 8.

Step 7: The question asks for x + y, so calculate: 2 + 8 = 10.

Step 8: This matches choice D.

The Trap: Choice A (4) is the most commonly picked wrong answer — students sometimes solve correctly for x = 2 but then mistakenly report a value related only to the coefficient in the first equation, or make an early substitution error that happens to produce 4 as a partial (incorrect) result.

POE — Eliminate the other three:

✗ A — 4 doesn't match either the correct x-value (2), y-value (8), or their sum (10) — likely a partial calculation error along the way.

✗ B — 6 doesn't correspond to a clean derivation from this system either.

✗ C — 8 is the correct value of y ALONE, not x + y — a student who correctly finds y = 8 but forgets to add x = 2 would mistakenly select this.

Solving with Desmos:

Step 1: Open Desmos. Step 2: Type the first equation exactly as given: y=4x

Step 3: Press Enter — Desmos graphs this line immediately.

Step 4: Type the second equation on the next line: 5x-3y=-14

Step 5: Press Enter — Desmos graphs this second line.

Step 6: Click directly on the point where the two lines intersect — Desmos will display the exact coordinates of that point, which will show (2, 8), meaning x = 2 and y = 8.

Step 7: Since the question asks for x + y, type a third line: 2+8 (or click and read both coordinates and add them yourself) — this gives 10, confirming choice D.

Systems of two linear equations in two variables — Q3 [Medium] (ID: 340afb12)

Question ID: 340afb12

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Systems of two linear equations in two variables	Medium

Question

Ellen bought two types of snack bags for a school event. Each box of chips Ellen bought contained 24 bags and cost $\$11$. Each box of crackers Ellen bought contained 16 bags and cost $\$9$. It cost Ellen $\$497$ before tax to buy 968 bags. If x is the number of boxes of chips and y is the number of boxes of crackers that Ellen bought, which of the following systems of equations represents this situation?

Answer

A. $11x + 16y = 968$

$24x + 9y = 497$

B. $11x + 9y = 968$

$24x + 16y = 497$

C. $11x + 9y = 497$

$24x + 16y = 968$

D. $9x + 11y = 497$

$16x + 24y = 968$

Correct Answer: C

The Rule: When translating a word problem into a system of equations, carefully match EACH number to what it's actually describing — cost-per-box goes with a cost total, and bags-per-box goes with a bags total. Mixing these up (using cost numbers in the bags equation, or vice versa) is the most common error in this question type.

The Tip: Make a small table before writing your equations: list each item (chips, crackers), its cost per box, its bags per box, and how many boxes were bought. This visual organization prevents the classic mistake of swapping which number belongs to which equation.

Why C is right: Chips cost $11 per box and contain 24 bags per box; crackers cost $9 per box and contain 16 bags per box. The COST equation is 11x + 9y = 497 (matching the total cost of $497). The BAGS equation is 24x + 16y = 968 (matching the total of 968 bags). This matches choice C exactly.

Step-by-Step Solution:

Step 1: Identify what x and y represent: x = number of boxes of chips, y = number of boxes of crackers.

Step 2: Build the COST equation using the price per box: chips cost $11 per box, crackers cost $9 per box, and the total cost was $497. This gives: 11x + 9y = 497.

Step 3: Build the BAGS equation using the bags per box: chips have 24 bags per box, crackers have 16 bags per box, and the total was 968 bags. This gives: 24x + 16y = 968.

Step 4: Compare these two equations to the answer choices — choice C lists exactly 11x + 9y = 497 and 24x + 16y = 968, matching perfectly.

The Trap: Choice B is the most commonly picked wrong answer — it swaps the numbers between the two equations, incorrectly pairing the bags-per-box numbers (24 and 16) with the cost total (968) and the cost-per-box numbers (11 and 9) with... actually it uses 11x+9y=968 and 24x+16y=497, which mismatches which total belongs to which set of per-box numbers.

POE — Eliminate the other three:

✗ **A —** This equation set pairs the wrong price ($9) with chips and the wrong price ($24, which is actually the bag count) with crackers in the cost equation — the numbers are scrambled between the two equations.

✗ **B —** This equation set swaps which TOTAL (968 bags vs. $497 cost) is paired with which set of per-box numbers — the cost equation incorrectly uses 968 and the bags equation incorrectly uses 497.

✗ **D —** This equation set swaps the roles of x and y entirely (putting crackers' numbers where chips' numbers should be, and vice versa) while also mismatching the totals — multiple errors combined.

Solving with Desmos:

Step 1: Open Desmos. Step 2: Type the cost equation you derived: 11x+9y=497

Step 3: Press Enter — Desmos graphs this line.

Step 4: Type the bags equation on the next line: 24x+16y=968

Step 5: Press Enter — Desmos graphs this second line.

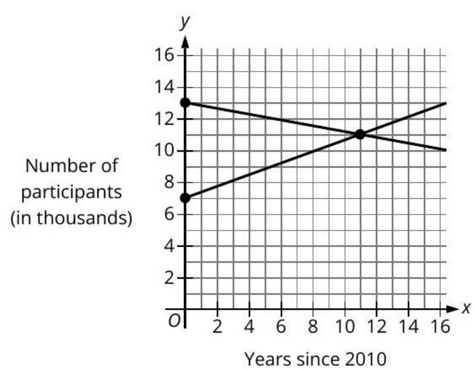
Step 6: Click on the intersection point of the two lines — Desmos will display the exact (x, y) coordinates, which should both be positive whole numbers (since you can't buy a fractional box) — this confirms your two equations correctly model the real-world situation, verifying choice C represents the system correctly, since it produces a sensible, whole-number solution.

Systems of two linear equations in two variables — Q4 **[Medium]** (ID: 080e75c2)

Question ID: 080e75c2

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Systems of two linear equations in two variables	Medium

Question



The functions f and g model the number of participants, in thousands, in two different programs x years since 2010. The graphs of $y = f(x)$ and $y = g(x)$ are shown. Which of the following could represent functions f and g?

Answer

A. $f(x) = -\frac{4}{11}x + 13$

$g(x) = \frac{2}{11}x + 7$

B. $f(x) = -\frac{4}{11}x + 7$

$g(x) = \frac{2}{11}x + 13$

C. $f(x) = -\frac{2}{11}x + 13$

$g(x) = \frac{4}{11}x + 7$

D. $f(x) = -\frac{2}{11}x + 7$

$g(x) = \frac{4}{11}x + 13$

Correct Answer: C

The Rule: To find the equation of a line from its graph, identify two clear points on the line, calculate the slope using (change in y) ÷ (change in x), then use the y-intercept (where the line crosses the y-axis) to write the full equation in slope-intercept form: $y = mx + b$.

The Tip: When a graph shows two lines and gives you their y-intercepts as clearly marked dots on the y-axis, start there — the y-intercept immediately tells you the 'b' value, narrowing down your answer choices before you even calculate the slope.

Why C is right: The decreasing line (f) starts at y-intercept 13 and passes through the intersection point near (11, 11); its slope is $(11 - 13)/(11 - 0) = -2/11$, giving $f(x) = -2/11\ x + 13$. The increasing line (g) starts at y-intercept 7 and passes through the same intersection point; its slope is $(11 - 7)/(11 - 0) = 4/11$, giving $g(x) = 4/11\ x + 7$. This matches choice C exactly.

Step-by-Step Solution:

Step 1: Identify the y-intercepts directly from the graph: the decreasing line crosses the y-axis at 13, and the increasing line crosses the y-axis at 7.

Step 2: Identify a second clear point on each line — both lines appear to cross paths at approximately (11, 11).

Step 3: Calculate the slope of the decreasing line using the two points (0, 13) and (11, 11): slope = $(11 - 13) / (11 - 0) = -2/11$.

Step 4: Calculate the slope of the increasing line using the two points (0, 7) and (11, 11): slope = $(11 - 7) / (11 - 0) = 4/11$.

Step 5: Write each equation in slope-intercept form: $f(x) = -2/11\ x + 13$ (the decreasing line) and $g(x) = 4/11\ x + 7$ (the increasing line).

Step 6: Compare to the answer choices — choice C lists $f(x) = -2/11x + 13$ and $g(x) = 4/11x + 7$, matching exactly.

The Trap: Choice D is the most commonly picked wrong answer — it swaps the y-intercepts between f and g (using 7 for f and 13 for g), which happens when a student correctly calculates both slopes but then mismatches which slope-intercept pair belongs to the increasing vs. decreasing line.

POE — Eliminate the other three:

✗ A — This uses the correct y-intercepts (13 for f, 7 for g) but the wrong slope fractions ($-4/11$ and $2/11$ instead of $-2/11$ and $4/11$) — the numerator and denominator relationship is reversed for both slopes.

✗ B — This uses the wrong y-intercepts (7 for f instead of 13, and 13 for g instead of 7) while also using the wrong slope magnitudes — multiple errors combined.

✗ D — This uses the correct slope magnitudes ($-2/11$ and $4/11$) but swaps which y-intercept (7 or 13) goes with which line, incorrectly pairing the decreasing line with intercept 7 and the increasing line with intercept 13.

Solving with Desmos:

Step 1: Open Desmos. Step 2: Since you're estimating a line from a picture, first type each answer choice's pair of equations as a test — start with choice C: y=-2/11x+13 and on the next line y=4/11x+7

Step 3: Press Enter after each — Desmos will graph both lines.

Step 4: Compare Desmos's graph to the original picture in the question: check that one line starts at (0, 13) and slopes downward, the other starts at (0, 7) and slopes upward, and that they cross near $x = 11$.

Step 5: If everything matches the original graph's shape and intercepts, you've confirmed choice C is correct. If you're unsure between two choices, graph both candidate pairs in different colors (Desmos automatically assigns different colors to each line) and see which pair visually matches the original picture's intercepts and steepness most closely.

Systems of two linear equations in two variables — Q5 **[Hard]** (ID: 59813abf)

Question ID: 59813abf

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Systems of two linear equations in two variables	Hard

Question

$$38(x-n)=38y+38n$$

One of the equations in a system of two linear equations is given, where n is a positive constant. The system has no solution. Which equation could be the second equation in this system?

Answer

A. $4x-4y=8n$

B. $4x+4y=4n$

C. $4x+4y=8n$

D. $4x-4y=4n$

Correct Answer: D

The Rule: A system of two linear equations has NO solution only when the two lines are PARALLEL but not identical — meaning they have the exact same slope (same ratio of x-coefficient to y-coefficient) but different y-intercepts (so they never touch).

The Tip: To quickly check whether two equations represent parallel, non-identical lines, first simplify both equations into the same basic form, then compare the ratio of their x and y coefficients. If those ratios match but the constant term doesn't scale the same way, you have a 'no solution' pair.

Why D is right: Simplify the given equation $38(x - n) = 38y + 38n$ by dividing every term by 38: $x - n = y + n$, which rearranges to $x - y = 2n$. For the system to have NO solution, the second equation must have the same left side pattern (same x and y coefficients) but a DIFFERENT constant relationship. Choice D, $4x - 4y = 4n$, simplifies (dividing by 4) to $x - y = n$. Since n is a positive constant, $n \neq 2n$ (they can only be equal if $n = 0$, but n is specified to be positive), so these two equations represent parallel lines with different intercepts — confirming no solution.

Step-by-Step Solution:

Step 1: Simplify the given equation: $38(x - n) = 38y + 38n$. Divide every term by 38: $x - n = y + n$.

Step 2: Rearrange into standard form: $x - y = n + n = 2n$.

Step 3: For a system to have NO solution, the second equation needs the exact same coefficient ratio for x and y (so the lines are parallel) but a DIFFERENT constant value relative to that same scaling (so the lines never actually meet).

Step 4: Test each answer choice by simplifying it to the same '$x - y = ...$' form. Choice D: $4x - 4y = 4n$. Divide by 4: $x - y = n$.

Step 5: Compare: the original equation says $x - y = 2n$, and choice D says $x - y = n$. Since n is a positive constant, 2n and n are always different values — confirming these are parallel, non-identical lines with NO solution.

Step 6: Double-check the other choices don't also produce this pattern (see POE below) — only choice D fits.

The Trap: Choice A ($4x - 4y = 8n$) is the most commonly picked wrong answer — dividing this by 4 gives $x - y = 2n$, which is IDENTICAL to the original simplified equation, meaning these two equations actually represent the exact SAME line, producing infinitely many solutions, not no solution.

POE — Eliminate the other three:

✗ A — Simplifying $4x - 4y = 8n$ by dividing by 4 gives $x - y = 2n$ — this is the exact same equation as the original ($x - y = 2n$), meaning these represent the SAME line, giving infinitely many solutions, not no solution.

✗ B — $4x + 4y = 4n$ has a different coefficient pattern ($+4y$ instead of $-4y$), meaning this line has a different (positive) slope than the original — different slopes mean the lines will cross at exactly one point, not zero.

✗ C — $4x + 4y = 8n$ also has a $+4y$ term, giving it a different slope than the original equation — again, different slopes mean exactly one solution, not no solution.

Solving with Desmos:

Step 1: Since this problem involves the unknown constant n, first pick a simple test value for n to make the equations concrete — let's use n = 1 (any positive number works the same way). Step 2: Open Desmos and type the original equation with n = 1 substituted in: 38(x-1)=38y+38

Step 3: Press Enter — Desmos graphs this line.

Step 4: Now type each answer choice with n = 1 substituted in, one at a time, starting with choice D: 4x-4y=4

Step 5: Press Enter — look at the graph. If the new line is perfectly parallel to (never touches) the first line, that choice gives NO solution, confirming it's correct.

Step 6: For comparison, also try choice A with n=1: 4x-4y=8 — you'll see this line lands EXACTLY on top of the original line (they're the same line), showing infinitely many solutions, which confirms choice A is wrong and helps you see the difference visually.

Systems of two linear equations in two variables — Q6 [Hard] (ID: 5e9b6079)

Question ID: 5e9b6079

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Systems of two linear equations in two variables	Hard

Question

$$y = 7x + 18$$

One of the equations in a system of two linear equations is given. The system has no solution. Which equation could be the second equation in the system?

Answer

A. $-21x + y = 54$

B. $-21x + y = 18$

C. $-7x + y = 21$

D. $-7x + y = 18$

Correct Answer: C

The Rule: For a system to have NO solution, the second equation's line must be PARALLEL to the first (same slope) but have a DIFFERENT y-intercept (so the lines run side by side forever without crossing).

The Tip: Rewrite the given equation into the standard form '(number)x + y = (number)' to easily compare its coefficients to each answer choice — matching coefficients with a different constant means no solution; matching coefficients AND constant means infinite solutions; different coefficients mean exactly one solution.

Why C is right: The given equation y = 7x + 18 can be rewritten as −7x + y = 18. For no solution, we need an equation with the same −7x + y pattern but a DIFFERENT constant than 18. Choice C, −7x + y = 21, has exactly this same coefficient pattern (−7 and 1) but a different constant (21 instead of 18), confirming these lines are parallel and never intersect.

Step-by-Step Solution:

Step 1: Rewrite the given equation y = 7x + 18 into standard form by subtracting 7x from both sides: −7x + y = 18.

Step 2: For no solution, look for an answer choice with the identical coefficients on x and y (−7 and 1) but a constant term different from 18.

Step 3: Check choice C: −7x + y = 21. The coefficients (−7 and 1) match exactly, and the constant (21) is different from 18 — this is the parallel-line pattern needed for no solution.

Step 4: Confirm this is correct and check the other choices don't also fit this pattern (see POE below).

The Trap: Choice D (−7x + y = 18) is the most commonly picked wrong answer — this has the exact same coefficients AND the exact same constant as the original equation, meaning it's not a second, different equation at all — it's the SAME line, giving infinitely many solutions instead of no solution.

POE — Eliminate the other three:

✗ **A —** −21x + y = 54 has a different x-coefficient (−21 instead of −7) than the original equation, meaning this line has a different, steeper slope — different slopes always produce exactly one solution, not zero.

✗ **B —** −21x + y = 18 also has a different x-coefficient (−21 instead of −7), giving it a different slope than the original line — this means exactly one solution, not no solution.

✗ **D —** −7x + y = 18 has identical coefficients AND an identical constant to the original equation — this isn't a different line at all, it's the exact same line, producing infinitely many solutions rather than none.

Solving with Desmos:

Step 1: Open Desmos and type the original equation: y=7x+18

Step 2: Press Enter — Desmos graphs this line.

Step 3: Type choice C on the next line, rearranged to match: y=7x+21 (this is the same as -7x+y=21, just solved for y to make graphing easier)

Step 4: Press Enter — look at the graph. You should see two lines that are perfectly parallel (same steepness, same direction) but shifted apart vertically, never touching no matter how far you zoom out — this visually confirms no solution.

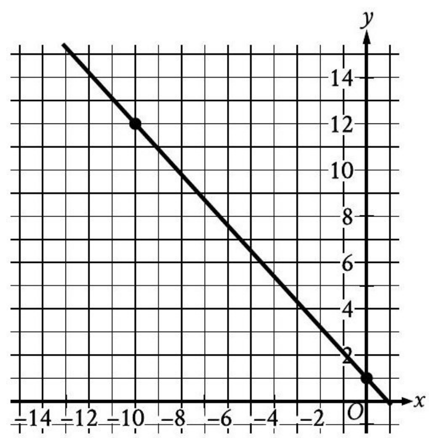
Step 5: To double check, try graphing choice D (y=7x+18) on a third line — you'll see it lands EXACTLY on top of the original line, since it's actually the identical equation, confirming why D gives infinitely many solutions instead of zero.

Systems of two linear equations in two variables — Q7 [Hard] [Grid-In] (ID: 18ac8354)

Question ID: 18ac8354

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Systems of two linear equations in two variables	Hard

Question



The graph of line g is shown in the *xy*-plane. Line k is defined by $165x + py = w$, where p and w are constants. If line k is graphed in this *xy*-plane, resulting in the graph of a system of two linear equations, the system of two linear equations will have infinitely many solutions. What is the value of $p + w$?

Correct Answer: 495 (Student-Produced Response — no answer choices given)

The Rule: For a system of two equations to have INFINITELY MANY solutions, the two equations must represent the EXACT SAME line — meaning every coefficient and constant in one equation is the same multiple of the corresponding value in the other equation.

The Tip: When you're given a graph and told a second equation (with unknown constants) must produce infinitely many solutions, first find the equation of the graphed line yourself in simple whole-number form, then figure out what single 'scale factor' turns your simple equation into the given equation's exact form.

Why this is the answer: From the graph, line g passes through (0, 2) and (−10, 12), giving a slope of (12 − 2)/(−10 − 0) = 10/(−10) = −1, so the line's equation is $y = -x + 2$, or equivalently $x + y = 2$. To make this match $165x + py = w$ exactly (for infinitely many solutions), multiply the entire equation $x + y = 2$ by 165: $165x + 165y = 330$. So $p = 165$ and $w = 330$, giving $p + w = 165 + 330 = 495$.

Step-by-Step Solution:

Step 1: Read two clear points off the graph of line g: it appears to pass through (0, 2) — the y-intercept — and (−10, 12).

Step 2: Calculate the slope using these two points: slope = (12 − 2) / (−10 − 0) = 10 / (−10) = −1.

Step 3: Write the equation of line g in slope-intercept form using the y-intercept (2) and slope (−1): $y = -x + 2$.

Step 4: Rearrange this into standard form to match the format of line k: $x + y = 2$.

Step 5: For line k ($165x + py = w$) to represent the exact same line as g ($x + y = 2$) — which is required for infinitely many solutions — every term must be scaled by the same factor. Since the x-coefficient goes from 1 to 165, the scale factor is 165.

Step 6: Apply this scale factor to the other terms: the y-coefficient becomes $1 \times 165 = 165$, so $p = 165$. The constant becomes $2 \times 165 = 330$, so $w = 330$.

Step 7: Calculate $p + w = 165 + 330 = 495$.

Step 8: This is a grid-in question — enter 495 as your final answer.

The Trap: A common error on this problem is misreading the graph's slope or y-intercept (for example, reading the y-intercept as 3 instead of 2, or misjudging the second point), which throws off the entire scale-factor calculation and produces a wrong final sum.

Solving with Desmos:

Step 1: Open Desmos. Step 2: To verify your reading of the graph, type y=-x+2 into Desmos and compare the resulting line to the picture given in the question — check that it passes through (0, 2) and (−10, 12) exactly as shown.

Step 3: Once confirmed, type the scaled version: 165x+165y=330 on a second line.

Step 4: Press Enter — Desmos should show this second line landing EXACTLY on top of the first line (they're the same line, just written differently), confirming that $p = 165$ and $w = 330$ make line k identical to line g.

Step 5: Calculate $p + w = 165 + 330 = 495$ and enter this into the grid-in box.

Linear functions

6 questions · Easy 1 / Medium 2 / Hard 3

Linear functions — Q1 **[Easy]** (ID: 0c490cd5)

Question ID: 0c490cd5

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear functions	Easy

Question

Sam has a flag with an area of 110 square inches (in^2). Sam plans to enlarge the flag by sewing on pieces of colored nylon, where each piece would increase the flag's area by $64\ \text{in}^2$. Which equation gives the flag's total area y, in square inches, after Sam sews on x pieces of nylon?

Answer

A. $y = 64x$

B. $y = 64 + 110x$

C. $y = 110 + 64x$

D. $y = 174x$

Correct Answer: C

The Rule: In a real-world 'starting amount plus additional amount per unit' scenario, the equation always takes the form: (total) = (starting amount) + (rate per unit)(number of units) — the starting amount stays fixed, and the rate is multiplied by however many units are added.

The Tip: Identify the 'starting point' (the value when x = 0) and the 'rate of change' (how much y increases for each additional unit of x) separately before writing the equation — the starting point is always the constant added alone, and the rate is always the number multiplied by x.

Why C is right: The flag starts with an area of 110 square inches (before any nylon pieces are added), so 110 is the fixed starting amount. Each piece of nylon adds 64 square inches, so 64 is the rate multiplied by x, the number of pieces added. This gives the equation y = 110 + 64x, matching choice C.

Step-by-Step Solution:

Step 1: Identify the fixed starting value: the flag begins with an area of 110 square inches before Sam adds any nylon pieces.

Step 2: Identify the rate of change: each nylon piece adds 64 square inches to the area.

Step 3: Identify the variable: x represents the number of nylon pieces added.

Step 4: Combine these into the equation: total area = starting area + (rate)(number of pieces), which gives y = 110 + 64x.

Step 5: This matches choice C.

The Trap: Choice B (y = 64 + 110x) is the most commonly picked wrong answer — it swaps which number is the starting amount and which is the rate, incorrectly treating 64 as the fixed starting area and 110 as the per-piece rate, when it's actually the reverse.

POE — Eliminate the other three:

✗ A — y = 64x completely omits the starting area of 110 square inches — this equation would incorrectly suggest the flag starts with zero area before any nylon is added.

✗ B — This swaps the roles of 110 and 64 — it uses 64 as the fixed starting amount and 110 as the rate per piece, which is backwards from what the problem describes.

✗ D — y = 174x incorrectly combines 110 and 64 into a single rate (174 = 110 + 64) and multiplies the entire sum by x, which doesn't correctly represent a fixed starting amount plus a separate per-unit rate.

Solving with Desmos:

Step 1: Open Desmos and type the equation you derived: y=110+64x

Step 2: Press Enter — Desmos will graph this line.

Step 3: Check the y-intercept by looking at where the line crosses the y-axis (or use the table feature to see the value when x = 0) — it should show y = 110, confirming the flag's starting area before any nylon is added.

Step 4: Use the table feature (click the '+' icon and select table, or add a table below the equation) and enter x = 1 — you should see y = 174, confirming that adding one piece of nylon increases the area by exactly 64 square inches (174 − 110 = 64), which matches the problem's description and confirms choice C.

Linear functions — Q2 **[Medium]** (ID: fc4a0d2a)

Question ID: fc4a0d2a

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear functions	Medium

Question

$$g(x) = 3(14x - 15)$$

What is the *y*-coordinate of the *y*-intercept of the graph of $y = g(x) - 2$ in the *xy*-plane?

Answer

A. -47

B. -45

C. -17

D. -15

Correct Answer: A

The Rule: The y-intercept of any function is the value of the function when x = 0. If a new function is defined as the original function shifted by a constant (like y = g(x) − 2), you find its y-intercept by first finding g(0), and then applying that same shift.

The Tip: Whenever you're asked about a graph like 'y = g(x) − 2' or 'y = g(x) + 5,' don't try to re-derive a whole new equation — just find the value of the ORIGINAL function at the point you care about (here, x = 0), and then simply add or subtract the shift amount at the very end.

Why A is right: First expand g(x) = 3(14x − 15) = 42x − 45. The y-intercept of g itself is g(0) = 42(0) − 45 = −45. Since we want the y-intercept of y = g(x) − 2, subtract 2 from g(0): −45 − 2 = −47.

Step-by-Step Solution:

Step 1: Expand the given function: g(x) = 3(14x − 15) = 3(14x) − 3(15) = 42x − 45.

Step 2: Find the y-intercept of g(x) itself by evaluating g(0): g(0) = 42(0) − 45 = −45.

Step 3: The question asks about the NEW function y = g(x) − 2, which shifts every output of g down by 2.

Step 4: Apply this shift to the y-intercept you just found: −45 − 2 = −47.

Step 5: This matches choice A.

The Trap: Choice C (−17) is the most commonly picked wrong answer — it typically results from a distribution error while expanding 3(14x − 15), such as computing 3 × 15 incorrectly or forgetting to distribute the 3 across BOTH terms inside the parentheses.

POE — Eliminate the other three:

✗ B — −45 is the y-intercept of the ORIGINAL function g(x) alone — this is the value BEFORE subtracting 2 for the shifted function y = g(x) − 2. A student who forgets to apply the final shift would stop here.

✗ C — −17 typically comes from a distribution error, such as computing 3(14x − 15) incorrectly and only partially multiplying through the parentheses.

✗ D — −15 is simply the constant that appears inside the original parentheses (14x − 15) — selecting this means mistaking a piece of the unexpanded expression for the final y-intercept, without doing any of the actual distribution or shifting work.

Solving with Desmos:

Step 1: Open Desmos and type the original function exactly as given: g(x)=3(14x-15)

Step 2: Press Enter — Desmos will graph this line and also let you use 'g(x)' in later expressions.

Step 3: On the next line, type the new shifted function: y=g(x)-2

Step 4: Press Enter — Desmos graphs this new line.

Step 5: To find its y-intercept, either look at where this new line crosses the y-axis on the graph, or click the wrench/settings icon and use the table feature: create a table, enter x = 0, and read the resulting y-value — it will show −47, confirming choice A.

Linear functions — Q3 **[Medium]** (ID: 2493c767)

Question ID: 2493c767

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear functions	Medium

Question

$$s(x) = 133 - 4x$$

The function s shown gives the number of spoons, $s(x)$, remaining in a dispenser x minutes after the dispenser had been loaded with spoons. Which statement is the best interpretation of $s(3) = 121$?

Answer

A. The number of spoons remaining in the dispenser decreased by a total of 3 spoons after 121 minutes.

B. There were 3 spoons remaining in the dispenser 121 minutes after the dispenser had been loaded with spoons.

C. There were 121 spoons remaining in the dispenser 3 minutes after the dispenser had been loaded with spoons.

D. The number of spoons remaining in the dispenser decreased by a total of 121 spoons after 3 minutes.

Correct Answer: C

The Rule: When a function s(x) models a real-world quantity over time, evaluating s(some number) = some result means: 'INPUT this number of [units] gives OUTPUT this amount of [quantity].' Always identify what the input variable represents and what the output represents before matching to an answer choice.

The Tip: Underline what x represents (here, minutes) and what s(x) represents (here, spoons remaining) directly in the problem before reading the answer choices — many wrong choices deliberately swap these two roles to test whether you're reading carefully.

Why C is right: In the function s(x) = 133 − 4x, x represents minutes after loading, and s(x) represents the number of spoons remaining. So s(3) = 121 means: when x = 3 (3 minutes have passed), s(x) = 121 (121 spoons remain in the dispenser).

Step-by-Step Solution:

Step 1: Identify what the input variable x represents in this context: minutes after the dispenser was loaded with spoons.

Step 2: Identify what the output s(x) represents: the number of spoons remaining in the dispenser at that time.

Step 3: Interpret s(3) = 121 by substituting these roles: the INPUT is 3 (meaning 3 minutes have passed), and the OUTPUT is 121 (meaning 121 spoons remain at that time).

Step 4: Match this interpretation to the answer choices: 'There were 121 spoons remaining in the dispenser 3 minutes after the dispenser had been loaded with spoons' is choice C.

The Trap: Choice B is the most commonly picked wrong answer — it swaps the input and output values, incorrectly stating there were '3 spoons remaining... 121 minutes' after loading, when it's actually the reverse: 121 spoons remaining after 3 minutes.

POE — Eliminate the other three:

✗ **A —** This choice invents a 'decrease' framing that doesn't match how function notation works — s(3) = 121 isn't describing a decrease of 3 spoons over 121 minutes; it's a direct input-output pair.

✗ **B —** This swaps which number is the input (minutes) and which is the output (spoons remaining) — it should be 121 spoons remaining after 3 minutes, not 3 spoons remaining after 121 minutes.

✗ **D —** This also swaps the input and output values (using 121 as if it were the time elapsed and 3 as if it were the spoon count), plus incorrectly frames it as a 'decrease' rather than a direct count remaining.

Solving with Desmos:

Step 1: Open Desmos and type the function exactly as given: s(x)=133-4x

Step 2: Press Enter — Desmos will graph this line and remember the function s(x) for later use.

Step 3: On the next line, type: s(3) directly — Desmos will instantly calculate and display the output value, showing 121.

Step 4: This confirms that when x = 3 (3 minutes), s(x) = 121 (121 spoons remain), matching the direct input-output interpretation in choice C. You can also click on the graphed line at x = 3 to visually see the point (3, 121) marked on the graph.

Linear functions — Q4 [Hard] (ID: d4572f55)

Question ID: d4572f55

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear functions	Hard

Question

x	y
-34	t
-17	$t+23$
0	$t+46$

For a linear relationship between x and y, the table gives three values of x and their corresponding values of y, where t is a constant. Which equation represents this relationship?

Answer

A. $y=-2x+t+23$

B. $y=2x+t+23$

C. $y=-\frac{23}{17}x+t+46$

D. $y=\frac{23}{17}x+t+46$

Correct Answer: D

The Rule: When a table gives you values that include an unknown constant (like t), find the slope using the differences between rows that DON'T involve solving for t directly — subtracting two rows that both contain 't' will cancel it out, leaving you with a clean numeric slope calculation.

The Tip: Look for two rows in the table where subtracting their y-values makes the unknown constant 't' cancel out completely — this lets you calculate the slope using only clean numbers, without ever needing to know what t actually equals.

Why D is right: Using the rows x = −17 (y = t + 23) and x = 0 (y = t + 46): the change in y is (t + 46) − (t + 23) = 23 (the t's cancel out), and the change in x is 0 − (−17) = 17. So the slope is 23/17. Using the y-intercept from the x = 0 row (y = t + 46), the equation is y = (23/17)x + t + 46, matching choice D.

Step-by-Step Solution:

Step 1: Choose two rows from the table where subtracting the y-values will cancel out the unknown t. Use x = −17 (y = t + 23) and x = 0 (y = t + 46).

Step 2: Calculate the change in y: (t + 46) − (t + 23) = t + 46 − t − 23 = 23 (the t cancels out completely).

Step 3: Calculate the change in x: 0 − (−17) = 17.

Step 4: Calculate the slope: (change in y) / (change in x) = 23/17.

Step 5: Since x = 0 gives y = t + 46 directly from the table, this row already tells you the y-intercept of the line: t + 46.

Step 6: Write the full equation in slope-intercept form: y = (23/17)x + t + 46.

Step 7: This matches choice D. As a check, verify with the third row (x = −34, y = t): plug x = −34 into the equation: y = (23/17)(−34) + t + 46 = 23(−2) + t + 46 = −46 + t + 46 = t. This matches the table exactly, confirming the equation is correct.

The Trap: Choice C is the most commonly picked wrong answer — it has the correct constant term (t + 46) but an incorrect slope with the fraction flipped and the wrong sign (−23/17 instead of the correct 23/17), likely from a sign error while calculating the change in y or change in x.

POE — Eliminate the other three:

✗ A — y = −2x + t + 23 uses a whole-number slope (−2) instead of the correct fraction (23/17), and pairs it with the wrong constant term (t + 23 instead of t + 46) — likely from using the wrong pair of rows or miscalculating entirely.

✗ B — y = 2x + t + 23 has the same issues as choice A but with a positive whole-number slope instead of negative — still doesn't match the correct fractional slope of 23/17.

✗ C — y = −23/17 x + t + 46 has the correct constant term but the wrong sign on the slope — the line should be increasing (positive slope), not decreasing, based on the table values.

Solving with Desmos:

Step 1: Since this table involves the unknown constant t, first pick a simple test value for t to make things concrete — let's use t = 0 for simplicity. Step 2: Open Desmos and create a table: click the '+' icon, select the table option, and enter the x-values from the problem (−34, −17, 0) in the x-column, and the corresponding y-values WITH t = 0 substituted (0, 23, 46) in the y-column.

Step 3: Desmos will plot these three points. Click the line-fitting or regression feature if available, or simply type y=mx+b below the table and adjust m and b using sliders until the line passes through all three points.

Step 4: You should find the line that fits is y = (23/17)x + 46 (when t = 0, this matches y = 23/17x + t + 46 exactly, since t + 46 = 0 + 46 = 46).

Step 5: This confirms the slope is 23/17 and the constant term follows the pattern t + 46, matching choice D.

Linear functions — Q5 [Hard] (ID: 1a2a0f59)

Question ID: 1a2a0f59

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear functions	Hard

Question

A yoga studio is offering a deal where the first session is free, the second session is half off the regular price, and the remaining sessions are at the regular price. If the regular price of a session is $\$25.40$, which function f gives the total cost, in dollars, of purchasing x sessions using this deal, where $x \geq 2$?

Answer

A. $f(x) = 25.40(x-1) + 12.70$

B. $f(x) = 25.40(x-2) + 12.70$

C. $f(x) = 25.40(x-1) + 12.70(x-2)$

D. $f(x) = 25.40(x-2) + 12.70(x-1)$

Correct Answer: B

The Rule: When a real-world pricing deal has DIFFERENT rules for the first few units and a STANDARD rule for all remaining units, build the equation by keeping the special-case amounts as separate fixed constants, and only multiply the standard rate by the units that follow the standard rule (not the total number of units).

The Tip: Carefully identify exactly which sessions get the SPECIAL treatment (free or discounted) and keep those as separate fixed dollar amounts — then multiply the regular price only by the number of REMAINING sessions (x minus however many special sessions there were), not by x itself.

Why B is right: The first session is free (\$0) and the second session is half off (\$25.40 / 2 = \$12.70) — these two special sessions contribute a fixed \$12.70 total (0 + 12.70), regardless of x. Every session from the 3rd session onward costs the full regular price of \$25.40, and there are $(x - 2)$ such sessions when $x \geq 2$. This gives the total cost function $f(x) = 25.40(x - 2) + 12.70$, matching choice B.

Step-by-Step Solution:

Step 1: Identify the special pricing for the first two sessions: session 1 is free (\$0), and session 2 is half price (\$25.40 ÷ 2 = \$12.70).

Step 2: Add these two special amounts together as a fixed constant: 0 + 12.70 = 12.70. This fixed amount doesn't change no matter how many total sessions (x) are purchased.

Step 3: Determine how many sessions are charged at the FULL regular price: since sessions 1 and 2 are special, all sessions from the 3rd onward are regular price — that's $(x - 2)$ sessions when x is the total number of sessions and $x \geq 2$.

Step 4: Build the full cost function: (fixed special-session cost) + (regular price) × (number of regular-price sessions) = $12.70 + 25.40(x - 2)$.

Step 5: This matches choice B: $f(x) = 25.40(x - 2) + 12.70$.

The Trap: Choice A is the most commonly picked wrong answer — it uses $(x - 1)$ instead of $(x - 2)$, which would be correct if only the FIRST session were special, but forgets that the SECOND session is ALSO discounted (not full price), so only sessions from the 3rd onward should be counted at full price.

POE — Eliminate the other three:

✗ **A —** Using $(x - 1)$ instead of $(x - 2)$ only accounts for the first session being free, but forgets that the second session is ALSO not at full price (it's half off) — this would overcharge for the second session.

✗ **C —** Multiplying both $(x - 1)$ and $(x - 2)$ together creates a nonsensical quadratic relationship — this doesn't match the simple linear 'fixed cost plus per-session rate' structure the problem describes.

✗ **D —** Similarly, multiplying $(x - 2)$ and $(x - 1)$ together creates an unnecessary quadratic term that doesn't match the linear pricing structure described in the problem.

Solving with Desmos:

Step 1: Open Desmos and type the function you derived: f(x)=25.40(x-2)+12.70

Step 2: Press Enter — Desmos graphs this line.

Step 3: Use the table feature to check specific values: click the '+' icon, add a table, and enter x = 2 (meaning exactly 2 sessions purchased). The output should show $f(2) = 25.40(0) + 12.70 = 12.70$, which makes sense — 2 sessions costing free + half-off = \$12.70 total, with no full-price sessions yet.

Step 4: Now enter x = 3 in the table — the output should jump to $f(3) = 25.40(1) + 12.70 = 38.10$, confirming that the 3rd session correctly adds one full \$25.40 charge on top of the \$12.70 base, verifying choice B is structured correctly.

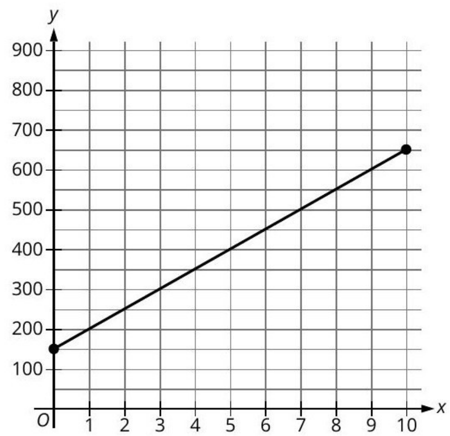
Linear functions — Q6 [Hard] [Grid-In] (ID: 590d662d)

Question ID: 590d662d

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear functions	Hard

Question

While walking on a trail, Mary stopped to read a trail sign. The graph shows the total distance y, in meters, Mary had walked on the trail x minutes after leaving the trail sign.



What distance, in meters, did Mary walk on the trail in the 10 minutes after leaving the trail sign?

Correct Answer: 500 (Student-Produced Response — no answer choices given)

The Rule: When a graph shows TOTAL accumulated distance over time, the question 'how much distance was covered DURING a specific time interval' is asking for the CHANGE in the y-value over that interval — not the final y-value itself. Always subtract the starting y-value from the ending y-value.

The Tip: Be very careful with wording like 'distance walked in the 10 minutes after' — this is NOT simply asking for the y-value at x = 10; it's asking for the DIFFERENCE between the y-value at the end of the interval and the y-value at the start of the interval.

Why this is the answer: From the graph, at x = 0 (right when Mary left the trail sign), she had already walked 150 meters total (the y-intercept). At x = 10 minutes later, she had walked 650 meters total. The distance walked DURING those 10 minutes is the difference: 650 − 150 = 500 meters.

Step-by-Step Solution:

Step 1: Read the starting point directly from the graph: at x = 0 minutes (right when Mary left the sign), the graph shows y = 150 meters already walked.

Step 2: Read the ending point from the graph: at x = 10 minutes, the graph shows y = 650 meters total walked.

Step 3: Since the question asks for the distance walked DURING the 10-minute interval (not the total distance at the 10-minute mark), calculate the CHANGE in y: 650 − 150 = 500.

Step 4: This is a grid-in question — enter 500 as your final answer, representing 500 meters walked during those 10 minutes.

The Trap: A common error on this problem is reading the final y-value (650) directly off the graph at x = 10 and submitting that as the answer, without realizing the question is asking for the CHANGE in distance during the interval, not the total accumulated distance at that point.

Solving with Desmos:

Step 1: Since this is a graph-reading problem, Desmos is most useful here for finding the exact equation of the line first. Estimate two clear points from the graph: (0, 150) and (10, 650). Step 2: Open Desmos and type these two points as a check: (0,150) and (10,650) on separate lines — Desmos will plot them as dots.

Step 3: Calculate the slope between them: type (650-150)/(10-0) on a new line — Desmos will compute this as 50, confirming Mary walks 50 meters per minute.

Step 4: To find the distance walked specifically during the 10-minute interval, type 650-150 directly — Desmos will show 500, confirming your answer. Enter 500 into the grid-in box.

Linear inequalities in one or two variables

3 questions · Easy 0 / Medium 3 / Hard 0

Linear inequalities in one or two variables — Q1 [Medium] (ID: 30b1e186)

Question ID: 30b1e186

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear inequalities in one or two variables	Medium

Question

In the *xy*-plane, which of the following does NOT contain any points that are part of the solution set to $5x - 7y > 35$?

Answer

A. The *x*-axis

B. The region where $x > 0$ and $y > 0$

C. The region where $x < 0$ and $y < 0$

D. The region where $x < 0$ and $y > 0$

Correct Answer: D

The Rule: To determine which region of the xy-plane satisfies a linear inequality, first rewrite the inequality in slope-intercept form (y < mx + b or y > mx + b) — this tells you immediately whether the solution region is ABOVE or BELOW the boundary line, which you can then compare against each answer choice's described region.

The Tip: After rewriting the inequality with y isolated, pick a few simple test points from each answer choice's described region (like the origin, or a point clearly in that quadrant) and plug them into the ORIGINAL inequality to quickly check whether that region could contain solution points.

Why D is right: Rewrite 5x − 7y > 35 by isolating y: −7y > 35 − 5x, then divide by −7 (flipping the inequality sign since we're dividing by a negative number): y < (5/7)x − 5. This means the solution region is entirely BELOW the line y = (5/7)x − 5. In the region where x < 0 and y > 0 (choice D), the line itself has already dropped below −5 for any negative x, meaning y would need to be even MORE negative than that to satisfy the inequality — but this region requires y > 0, which can never be low enough. So this region contains no solution points.

Step-by-Step Solution:

Step 1: Start with the given inequality: 5x − 7y > 35.

Step 2: Isolate y by first subtracting 5x from both sides: −7y > 35 − 5x, or equivalently −7y > −5x + 35.

Step 3: Divide both sides by −7. Remember: dividing an inequality by a NEGATIVE number flips the inequality sign. This gives: y < (5/7)x − 5.

Step 4: This tells you the solution set is every point BELOW the line y = (5/7)x − 5.

Step 5: Test choice D's region (x < 0 and y > 0): for any negative x-value, the boundary line y = (5/7)x − 5 is already less than −5 (since (5/7) times a negative number is negative, minus 5 makes it even more negative). For a point to satisfy the inequality, y would need to be even smaller (more negative) than this already-negative boundary value.

Step 6: But choice D requires y > 0 (positive), which can never be less than a negative boundary value — so this region contains zero solution points, confirming choice D is correct.

The Trap: Choice C is the most commonly picked wrong answer — students sometimes assume that because BOTH x and y are negative, the inequality (which involves subtraction) can't be satisfied, but testing an actual point (like x = −1, y

= −100) shows 5(−1) − 7(−100) = −5 + 700 = 695, which IS greater than 35 — so this region actually DOES contain solution points.

POE — Eliminate the other three:

✗ **A —** The x-axis (where y = 0) does contain solution points — for example, at x = 10, y = 0: 5(10) − 7(0) = 50, which is greater than 35, satisfying the inequality.

✗ **B —** The region where x > 0 and y > 0 does contain solution points — for example, at x = 100, y = 1: 5(100) − 7(1) = 500 − 7 = 493, which is greater than 35.

✗ **C —** The region where x < 0 and y < 0 does contain solution points — for example, at x = −1, y = −100: 5(−1) − 7(−100) = −5 + 700 = 695, which is greater than 35.

Solving with Desmos:

Step 1: Open Desmos and type the inequality exactly as given: 5x-7y>35

Step 2: Press Enter — Desmos will shade the entire region of the plane that satisfies this inequality (this is one of Desmos's most useful features for inequality problems — it does the region-testing for you visually).

Step 3: Look specifically at the upper-left area of the graph (where x < 0 and y > 0, matching choice D's description) — you'll see this entire area is UNSHADED, meaning no points there satisfy the inequality.

Step 4: For comparison, check the other three regions described in choices A, B, and C — you'll see the shaded (solution) region does extend into those areas, confirming they DO contain solution points, while only the upper-left region (choice D) is completely free of shading.

Linear inequalities in one or two variables — Q2 [Medium] (ID: d914abec)

Question ID: d914abec

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear inequalities in one or two variables	Medium

Question

To complete a landscaping project, Adam charges a fee of $\$20.00$ for his equipment and $\$9.50$ per hour spent working on the project. To complete this same landscaping project, Caroline charges a fee of $\$17.00$ for her equipment and $\$10.00$ per hour spent working on the project. If x represents the number of hours spent working on the landscaping project, what are all the values of x for which Caroline's total charge is greater than Adam's total charge?

Answer

A. $5 \leq x \leq 6$

B. $6 \leq x \leq 7$

C. $x < 5$

D. $x > 6$

Correct Answer: D

The Rule: To compare two people's or companies' pricing structures and find when one is greater than the other, set up an expression for each total charge, then write an inequality with one total GREATER THAN the other exactly as described in the question, and solve for the variable.

The Tip: Write out each person's total charge as its own separate expression first (fixed fee + rate × hours), clearly labeled, before combining them into an inequality — this prevents mixing up whose fee and whose rate belong in which part of the inequality.

Why D is right: Caroline's total charge is 17 + 10x (her $17 fee plus $10 per hour). Adam's total charge is 20 + 9.5x (his $20 fee plus $9.50 per hour). The question asks when Caroline's charge is GREATER than Adam's: 17 + 10x > 20 + 9.5x. Subtracting 9.5x from both sides and 17 from both sides gives 0.5x > 3, so x > 6.

Step-by-Step Solution:

Step 1: Write Caroline's total charge as an expression: her fee is \$17, plus \$10 for every hour (x), giving $17 + 10x$.

Step 2: Write Adam's total charge as an expression: his fee is \$20, plus \$9.50 for every hour (x), giving $20 + 9.5x$.

Step 3: Set up the inequality for when Caroline's charge is GREATER than Adam's: $17 + 10x > 20 + 9.5x$.

Step 4: Subtract $9.5x$ from both sides to gather the x-terms on one side: $17 + 0.5x > 20$.

Step 5: Subtract 17 from both sides: $0.5x > 3$.

Step 6: Divide both sides by 0.5: $x > 6$.

Step 7: This matches choice D.

The Trap: Choice C ($x < 5$) is the most commonly picked wrong answer — it results from setting up the inequality backwards (solving for when ADAM's charge is greater than Caroline's instead of the reverse) combined with an arithmetic slip, flipping both the inequality direction and the resulting boundary number.

POE — Eliminate the other three:

✗ A — $5 \le x \le 6$ introduces an upper bound that doesn't exist in this problem — there's no reason x would be capped at 6; the inequality is simply $x > 6$ with no upper limit.

✗ B — $6 \le x \le 7$ similarly introduces a nonexistent upper bound, and the boundary values don't align with the correct solution $x > 6$.

✗ C — $x < 5$ reverses the direction of the inequality (suggesting Caroline's charge is greater when x is SMALL, not large) and uses an incorrect boundary number — this would actually represent when Adam's charge exceeds Caroline's, not the reverse.

Solving with Desmos:

Step 1: Open Desmos and type Caroline's total charge as a function: c(x)=17+10x

Step 2: Press Enter, then type Adam's total charge on the next line: a(x)=20+9.5x

Step 3: Press Enter — Desmos graphs both lines. Look for the point where they cross (intersect) — click on it to see its exact coordinates, which should show $x = 6$.

Step 4: To confirm which side of $x = 6$ has Caroline's charge GREATER than Adam's, look at the graph to the RIGHT of $x = 6$ (larger x-values) — you'll see Caroline's line (c(x)) is now ABOVE Adam's line (a(x)), confirming that for $x > 6$, Caroline charges more, matching choice D.

Step 5: As an extra check, type c(x)>a(x) directly into Desmos on a new line — Desmos will shade the x-values where this is true, visually confirming the region is $x > 6$.

Linear inequalities in one or two variables — Q3 [Medium] (ID: e8fd6d99)

Question ID: e8fd6d99

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear inequalities in one or two variables	Medium

Question

Terry has a goal to collect at least 16 items per week for a donation drive. This week, Terry has collected 4 items. If x represents the additional number of items Terry needs to collect this week to meet the goal, which inequality represents this situation?

Answer

A. $4 - x \le 16$

B. $4 + x \le 16$

C. $4 - x \ge 16$

D. $4 + x \ge 16$

Correct Answer: D

The Rule: Phrases like 'at least' in a word problem always translate to a 'greater than or equal to' (≥) inequality symbol, never a strict 'greater than' (>) — because 'at least 16' includes the possibility of exactly reaching 16, not just exceeding it.

The Tip: Build a mental translation table for common inequality phrases before setting up any word problem: 'at least' means ≥, 'at most' means ≤, 'more than' means strictly >, and 'fewer than' means strictly <. Misreading these phrases is one of the most common sources of errors on inequality word problems.

Why D is right: Terry has already collected 4 items and needs x more items, so the total collected will be $4 + x$. Since the goal is to collect 'at least 16 items,' this total must be greater than or equal to 16, giving the inequality $4 + x \geq 16$.

Step-by-Step Solution:

Step 1: Identify the two quantities being added together: the 4 items Terry has already collected, and x, the additional items still needed.

Step 2: Write an expression for the total number of items Terry will have collected by the end of the week: $4 + x$.

Step 3: Translate the phrase 'at least 16 items' into an inequality symbol: 'at least' means the total must be greater than OR EQUAL TO 16, so use the ≥ symbol.

Step 4: Combine these pieces into the final inequality: $4 + x \geq 16$.

Step 5: This matches choice D.

The Trap: Choice B ($4 + x \leq 16$) is the most commonly picked wrong answer — it uses the correct expression ($4 + x$) but the WRONG inequality direction, mistakenly treating the goal as an upper limit ('at most 16') instead of a minimum threshold ('at least 16').

POE — Eliminate the other three:

✗ A — $4 - x \leq 16$ incorrectly subtracts x instead of adding it, which would represent items being taken away rather than items being collected — this doesn't match the situation of Terry ADDING more items toward the goal.

✗ B — $4 + x \leq 16$ uses the correct addition but the wrong inequality direction — 'at least' means ≥ (a minimum), not ≤ (a maximum), so this reverses the intended meaning of the goal.

✗ C — $4 - x \geq 16$ incorrectly subtracts x AND uses it in a way that would require x to be a large negative number to ever reach 16, which doesn't logically fit collecting additional items toward a goal.

Solving with Desmos:

Step 1: Open Desmos and type the inequality you derived: 4+x>=16 (in Desmos, type the greater-than-or-equal-to symbol by typing the '>' key followed by '=', and Desmos will automatically convert it to the ≥ symbol on screen).

Step 2: Press Enter — Desmos will display a shaded region or highlighted portion of the number line (since this is a one-variable inequality) showing all values of x that make the inequality true.

Step 3: You should see the shading starts at $x = 12$ and continues to the right (toward positive infinity) with a solid/filled boundary point at $x = 12$ (the filled-in point confirms $x = 12$ itself IS included, matching the 'or equal to' part of ≥).

Step 4: This confirms that Terry needs to collect AT LEAST 12 more items ($x \geq 12$) to reach the goal of 16 total, and the inequality $4 + x \geq 16$ correctly models this, matching choice D.